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Applicable Codes and Standards in Electrical & Electronic Engineering

IEEE, IEC, and ISO in Professional Engineering Practice

Learning Outcomes

- Define engineering codes and standards
- Explain why standards are important in professional practice
- Differentiate between code and standard
- Understand the role of IEEE, IEC, and ISO
- Identify important electrical and electronics standards
- Explain how standards improve safety and reliability
- Connect standards with engineering ethics and social responsibility

What is a Code?

Codes are **mandatory rules or regulations** created by governments, regulatory bodies, or industry authorities to protect public safety, health, and welfare.

They tell engineers **what must be followed by law.**

Examples:

- National Electrical Code (electrical wiring rules)
- BNBC

Example: When wiring a building:

The electrical code may require:

- Proper grounding
- Circuit breaker protection
- Minimum wire size
- Safe installation distance

If you violate the code, the building may fail inspection or become illegal to operate.

Key points of Codes:

- ✓ Mandatory
- ✓ Legally enforceable
- ✓ Focus strongly on public safety
- ✓ Violating them can lead to penalties or legal issues

What is a Standard?

Standards provide **recommended technical rules, specifications, and procedures** for designing, manufacturing, testing, and operating systems safely and efficiently.

They tell engineers **how to do something properly.**

Examples:

IEEE standard for power quality

IEC standard for transformer testing

ISO 9001 for quality management

Example: Suppose you are designing a solar inverter. A standard may specify:

- Voltage limits
- Efficiency requirements
- Testing procedure
- Safety performance

This ensures products from different companies work reliably.

Why Do Engineers Need Standards?

Without standards, every engineer might design differently.

That creates chaos.

Imagine if each company used different:

- Voltage levels
- Plug shapes
- Communication protocols
- Battery charging methods

Problems:

- Equipment incompatibility
- Safety hazards
- High failure rate

Example: Phone Charger

Why can chargers from different companies often work with many devices?

Because manufacturers follow standards.

Without standards:

- One charger for each device
- Unsafe charging
- Device damage

Standards and Engineering Ethics

Professional engineers have ethical duties.

Important canon: Protect public health, safety, and welfare.

Following standards is part of ethical engineering.

Ignoring standards can mean:

- Negligence
- Poor professional judgment
- Unsafe design

IEEE Standards

What is IEEE?

IEEE stands for: **Institute of Electrical and Electronics Engineers**

It is one of the largest engineering organizations in the world.

It develops standards for modern technology. Fields include:

- Power systems
- Electronics
- Communication
- AI
- Robotics
- Smart grid
- Renewable energy

Why IEEE Standards Matter

IEEE standards ensure equipment from different manufacturers can work together safely. They also improve:

- Reliability
- Safety
- Interoperability
- Efficiency

Important IEEE Standards

IEEE 1547: Interconnection of Distributed Energy Resources

Very important for:

- Solar PV
- Wind turbines
- Battery storage
- Microgrids

This standard defines how DER should connect to utility grids safely.

Example: A rooftop solar system injects power into grid.

Problems without standard:

- Voltage rise
- Reverse power flow
- Protection failure

IEEE 1547 prevents these.

Why Important in Bangladesh?

Increasing rooftop solar and battery storage makes grid integration standards important in Bangladesh.

Important IEEE Standards

IEEE 519

Harmonic Control

Modern power electronics create harmonics.

Examples:

- Inverters
- UPS
- EV chargers
- Variable speed drives

What are Harmonics?

In ideal power systems:

Current waveform = smooth sine wave

Harmonics distort waveform.

Effects:

- Overheating
- Transformer loss
- Motor vibration
- Poor power quality

IEEE 519 sets acceptable harmonic limits.

Important IEEE Standards

IEEE 802 Family

Includes:

IEEE 802.3

Ethernet

IEEE 802.11

Wi-Fi

Used in:

- Routers
- Laptops
- Phones
- IoT devices

Example

Why does Wi-Fi work between devices of different brands?

Because they follow IEEE 802.11.

Important IEEE Standards

IEEE 80: Substation Grounding

Used in power engineering.

Helps design safe grounding systems.

Protects workers from:

Electric shock

Example

During fault current in substation:

Ground potential rises.

Improper grounding may kill workers.

IEEE 80 helps prevent this.

IEC Standards

What is IEC?

IEC stands for: **International Electrotechnical Commission**

IEC prepares international standards for electrical and electronic systems.

Its goal: Ensure electrical products are safe and reliable worldwide.

Why IEC Matters

IEC standards are used in:

- Cables
- Batteries
- Switchgear
- Transformers
- Relays
- Electronics

They allow global compatibility.

Important IEC Standards

IEC 61850

Communication Networks for Substations

Used in:

- Digital substations
- Smart grid
- Automation systems

Allows communication among:

- Protection relays
- Breakers
- SCADA
- Controllers

Why Important? Modern substations are becoming intelligent.

Instead of only copper wiring, devices communicate digitally.

Example: Fault occurs.

Relay sends digital trip command to breaker. Breaker disconnects faulty section quickly.

This improves:

- Reliability
- Speed
- Safety

Important IEC Standards

IEC 61000

Electromagnetic Compatibility (EMC)

Controls electromagnetic interference.
Important in electronics.

Example: Suppose your inverter interferes with nearby communication devices.

Possible issues:

- Signal noise
- Data corruption
- Device malfunction

IEC 61000 reduces these problems.

IEC 62133

Battery Safety

Used for:

- Lithium-ion batteries
- EV batteries
- Portable devices
- Energy storage systems

Why Important? Battery failures can cause:

- Fire
- Thermal runaway
- Explosion

IEC 62133 helps ensure battery safety.

ISO Standards

What is ISO?

ISO stands for: **International Organization for Standardization**

ISO develops standards for organizations and management systems.

Unlike IEEE or IEC, ISO often focuses on processes.

Why ISO Matters for Engineers

Engineering is not only technical design.

It also includes:

- Quality management
- Documentation
- Safety culture
- Environmental responsibility

Important ISO Standards

ISO 9001 Quality Management System

Focus: Deliver consistent quality.
Organizations using ISO 9001
ensure:

- Better documentation
- Better testing
- Better quality control

Example: Transformer manufacturing company.

Without ISO 9001:

- Random testing
- Inconsistent quality

With ISO 9001:

- Standard process
- Fewer defects
- Better reliability

Important ISO Standards

ISO 14001

Environmental Management System

Important for sustainable engineering.

Helps organizations reduce:

- Pollution
- Waste
- Carbon emissions

Example: Power plant implementing ISO 14001 may reduce:

- Water waste
- Air pollution
- Hazardous waste

This supports sustainability.

Important ISO Standards

ISO 45001

Occupational Health and Safety

Focus: Worker safety.

Important in:

- Construction sites
- Factories
- Power plants
- Electrical maintenance

Example: High-voltage maintenance work.

Hazards:

- Arc flash
- Electric shock
- Fall from height

ISO 45001 improves safety procedures.

Combined Example: Solar Power Plant

Suppose engineers build a utility-scale solar farm.
Many standards apply simultaneously.

IEEE

Grid interconnection

IEC

PV modules, inverter safety, cables

ISO

Quality, safety, environment

This shows real engineering requires multiple standards.